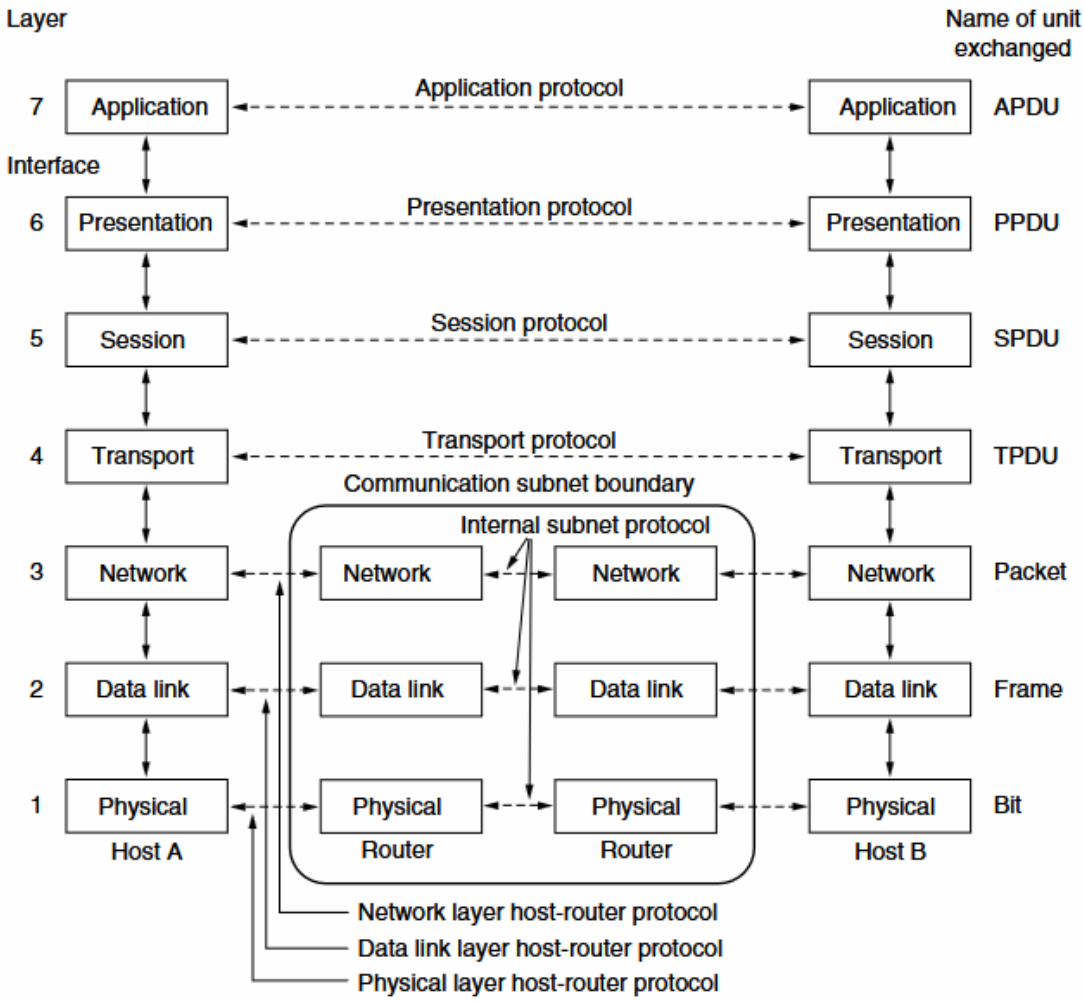
The OSI Reference Model and TCP/IP Model

The OSI Reference Model :

The OSI (Open Systems Interconnection) Reference Model was developed by the International Organization for Standardization (ISO) to standardize network communication between different computer systems. It provides a common framework that enables devices from different manufacturers to communicate effectively. The OSI model consists of seven layers, each performing a specific function in the data communication process.



Principles of the OSI Model

1. Create a layer only when a different level of abstraction is required.
2. Each layer should perform a specific and well-defined function.
3. Functions of each layer should support internationally standardized protocols.
4. Layer boundaries should minimize the exchange of information between layers.
5. The number of layers should be large enough to separate different functions but small enough to keep the architecture simple and manageable.

Layer	Name	Summary
Layer 7	**Application Layer**	Provides network services directly to end-user applications such as

		web browsers, email, and file transfer. It acts as the interface between the user and the network.
Layer 6	**Presentation Layer**	Translates, encrypts, decrypts, compresses, and formats data so that it can be understood by the receiving system.
Layer 5	**Session Layer**	Establishes, manages, synchronizes, and terminates communication sessions between applications running on different devices.
Layer 4	**Transport Layer**	Ensures reliable end-to-end communication by performing segmentation, error detection, flow control, and data recovery. It uses protocols such as TCP and UDP.
Layer 3	**Network Layer**	Determines the best path for data transmission and performs logical addressing and routing using IP addresses. Routers operate at this layer.
Layer 2	**Data Link Layer**	Provides reliable node-to-node communication by framing data, detecting errors, and controlling access to the transmission medium. Switches operate at this layer.
Layer 1	**Physical Layer**	Transmits raw bits over the physical medium using electrical, optical, or radio signals. It defines cables, connectors, voltages, and transmission speeds.

TCP/IP Reference Model

The **TCP/IP (Transmission Control Protocol/Internet Protocol) Reference Model** was developed for the **ARPANET**, the predecessor of the Internet, and later became the standard architecture for the Internet. It was designed by **Vinton Cerf and Robert Kahn** to enable communication across multiple interconnected networks. The model emphasizes **interoperability, scalability, and reliability**, ensuring that communication continues even if some network devices or transmission links fail. Unlike the OSI model, the TCP/IP model has **four layers**: Link, Internet, Transport, and Application.

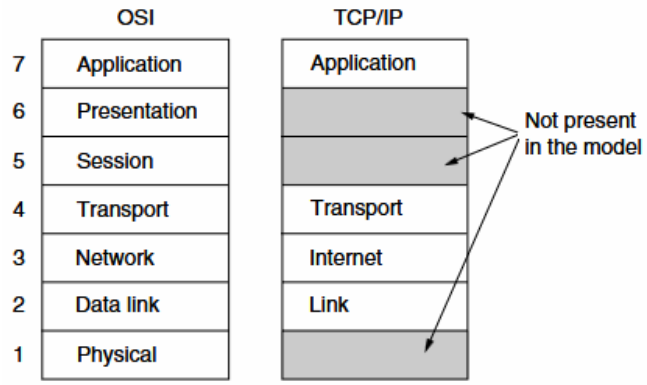


Figure 1-21. The TCP/IP reference model.

1. Link Layer

The **Link Layer** is the lowest layer of the TCP/IP model and is responsible for transmitting data over the physical network. It defines how data is sent across different network technologies such as **Ethernet, Wi-Fi, and serial links**. It handles framing, physical addressing, error detection, and communication between a host and the transmission medium.

2. Internet Layer

The **Internet Layer** is responsible for **logical addressing and routing** of packets across multiple interconnected networks. It uses the **Internet Protocol (IP)** to ensure that packets travel independently from the source to the destination, even if they take different paths. Since packets may arrive out of order, higher layers handle packet sequencing and reassembly if required.

3. Transport Layer

The **Transport Layer** provides **end-to-end communication** between applications running on different hosts. It is responsible for segmentation, error detection, flow control, and reliable data delivery. The two main protocols at this layer are **TCP (Transmission Control Protocol)**, which provides reliable communication, and **UDP (User Datagram Protocol)**, which offers faster but connectionless communication.

4. Application Layer

The **Application Layer** provides network services directly to user applications. It combines the functions of the **Application, Presentation, and Session layers** of the OSI model. It supports protocols such as **HTTP** (web browsing), **FTP** (file transfer), **SMTP** (email), **DNS** (domain name resolution), and **Telnet** (remote login), enabling users to access various Internet services.

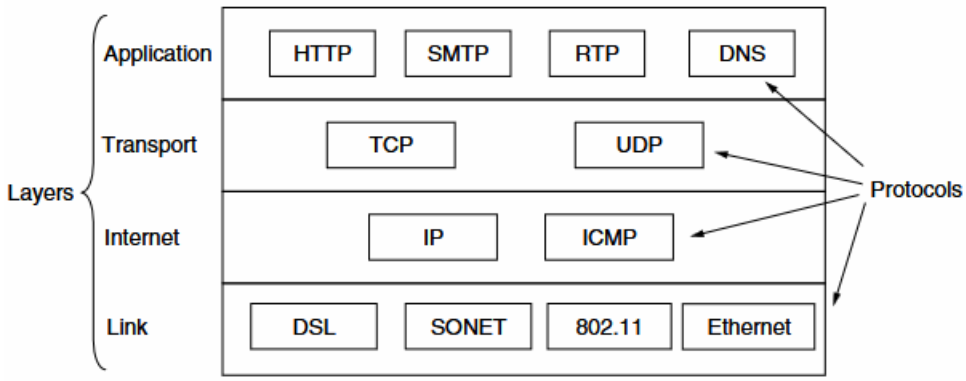


Figure 1-22. The TCP/IP model with some protocols we will study.

Model used in the course

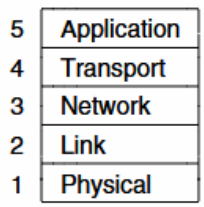


Figure 1-23. The reference model used in this book.

Comparison between OSI and TCP/IP Reference Models

The **OSI** and **TCP/IP** reference models are widely used networking models that provide a layered approach to network communication. Both models use independent protocol layers and provide **end-to-end communication**, but they differ in their design, number of layers, and implementation.

Similarities

- Both use a **layered architecture**.
- Both provide **end-to-end communication**.
- Both have **Application, Transport, and Network (Internet)** layers with similar functions.
- Both support communication between different computer systems.

OSI Model	TCP/IP Model
Developed by **ISO (International Organization for Standardization)**.	Developed by the **U.S. Department of Defense (DoD)** for ARPANET.
Consists of **7 layers**.	Consists of **4 layers**.
Clearly separates **Services, Interfaces, and Protocols**.	Does not clearly distinguish between services, interfaces, and protocols.
The **reference model was developed before the protocols**.	**Protocols were developed first**, and the model was created later.
More **general and protocol-independent**.	Specifically designed for the **TCP/IP protocol suite**.
Supports **connection-oriented and connectionless** communication in the Network layer, and **connection-oriented** communication in the Transport layer.	Supports **connectionless** communication in the Internet layer and both **connection-oriented (TCP)** and **connectionless (UDP)** communication in the Transport layer.
Mainly used as a **reference and teaching model**.	Used as the **standard model for the Internet**.

Key Concepts of the OSI Model

The OSI model is based on three important concepts:

1. **Services** – Define what each layer provides to the layer above it.
2. **Interfaces** – Specify how the upper layer accesses the services.
3. **Protocols** – Define the rules used for communication between corresponding layers.

Conclusion

The **OSI model** is a **theoretical reference model** that clearly separates services, interfaces, and protocols, making it useful for learning and network design. The **TCP/IP model** is a **practical implementation model** used by the Internet and focuses on efficient and reliable communication across interconnected networks. Despite their differences, both models provide a structured approach to network communication.